7(a)(i)	$^{210}_{83}Bi \rightarrow {}^{0}_{-1}e + {}^{210}_{84}Po$ + (anti) neutrino Accept beta particle	**B2**
(a)(ii)	total momentum/ total mass-energy/ total nucleon no/ total charge. Any two.	**A2**
(b)(i)	Mass of protons + neutrons = [(83 x 1.00729) + (127 x 1.00867)] u = 211.706 u Mass defect = 211.706 u - 209.939 u Mass defect = 1.767 u	**M1** **M1** **A0**

Qns	Answer	Marks
(b)(ii)	Binding energy = $1.767 \times 1.66 \times 10^{-27} \times (3.00 \times 10^{8})^{2}$ $= 2.64 \times 10^{-10}$ J Binding energy per nucleon = $[2.64 \times 10^{-10} / 1.60 \times 10^{-19} \times 10^{6}]/210$ = 7.86 MeV	M1 A1